

Converting PDF to XML

1. Introduction

Born-digital documents keep their text as real characters, so a converter can read the exact font, size and colour of every word. The hard part is recovering structure: which lines form a paragraph, where a heading starts, and in which order the columns are read. This sentence has **bold words**, *italic words* and a **red warning**. Water is H₂O and energy is $E = mc^2$.

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1.1 Goals

- Keep every character of the text layer.
- Keep styles: font, size, colour, bold and italic.
- Produce valid XML, EPUB and DOCX.

2. Method

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1. Extract words with their styles.
2. Detect blocks and reading order.
3. Merge both layers by position.

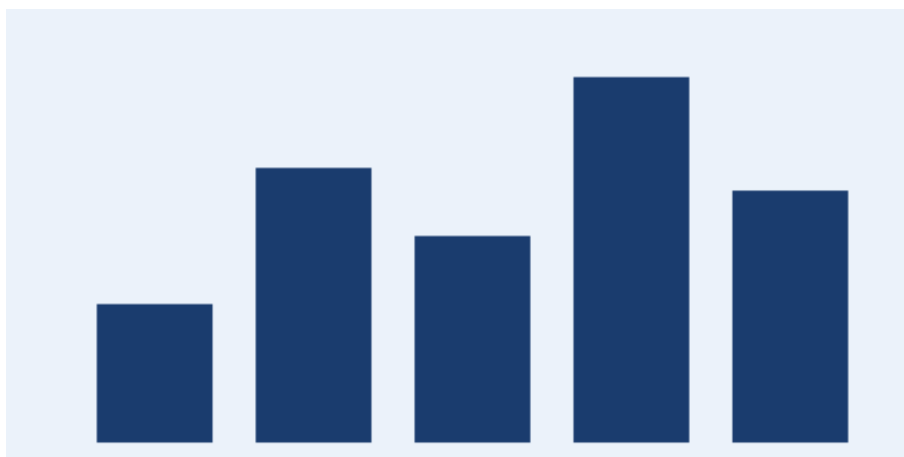


Figure 1. Characters kept per stage.

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